Problem Statement Document: Credit Risk IDSS

1. Problem Statement

The real-world problem this Intelligent Decision Support System (IDSS) addresses is the accurate prediction of credit risk and loan default for financial institutions. Currently, lenders face a continuous trade-off: approving risky loans leads to severe financial losses through defaults, while overly conservative lending criteria result in lost revenue and market share from rejecting reliable customers. This IDSS aims to analyze an applicant's financial history, demographic profile, and requested loan specifics to predict the likelihood of default before a loan is issued.

2. Stakeholders and Decision-Making Needs

The primary stakeholders who will utilize this system and benefit from its output include:

- **Loan Officers and Underwriters:** These front-line employees need a reliable, data-driven recommendation on whether to approve or reject an individual loan application. They also require an interpretable risk score (and an understanding of the key features driving that score) to justify their lending decisions to both applicants and management.
- **Credit Risk Managers:** These stakeholders require aggregate portfolio risk insights to adjust broader lending policies, calibrate interest rates, and refine internal loan grade parameters (especially in middle-tier grades like B, C, and D, where human judgment is most uncertain).

3. Business Value

Implementing this IDSS provides significant, quantifiable business value by:

- **Minimizing Financial Loss:** Accurately identifying high-risk applicants reduces the default rate and associated bad debt. Given that nearly 21.9% of historical loans in the dataset resulted in default, identifying these before approval will save substantial capital.
- **Increasing Revenue:** The system allows the institution to safely approve borderline "good" loans that manual underwriting might mistakenly reject, capturing additional interest revenue.
- **Improving Operational Efficiency:** Automating the initial screening and risk-scoring of applications accelerates the approval process, improving customer satisfaction and reducing manual labor costs.

4. IDSS Objective

Task Definition: This is a Binary Classification task. The system will predict the loan_status feature, where 0 represents a non-default (a safe loan) and 1 represents a default (a risky loan).

5. Evaluation Metrics

Because our exploratory data analysis revealed a significant class imbalance (roughly 78.1% non-defaults vs. 21.9% defaults), standard Accuracy will be a misleading metric. Therefore, the IDSS will be evaluated using the following metrics:

- **Recall (Sensitivity):** This is our most critical metric. In credit risk, a False Negative (approving a loan that ends up defaulting) is vastly more expensive than a False Positive (rejecting a safe loan). High recall ensures the model successfully catches the majority of actual defaults.
- **Precision and F1-Score:** While recall is prioritized, Precision is still important to ensure we do not indiscriminately reject good customers, which would harm revenue. The F1-Score will provide a balanced measure between Precision and Recall.
- **ROC-AUC:** This metric will be used to evaluate the model's overall ability to distinguish between the default and non-default classes across various probability thresholds, ensuring the model is robust regardless of the specific cutoff point chosen by the business.